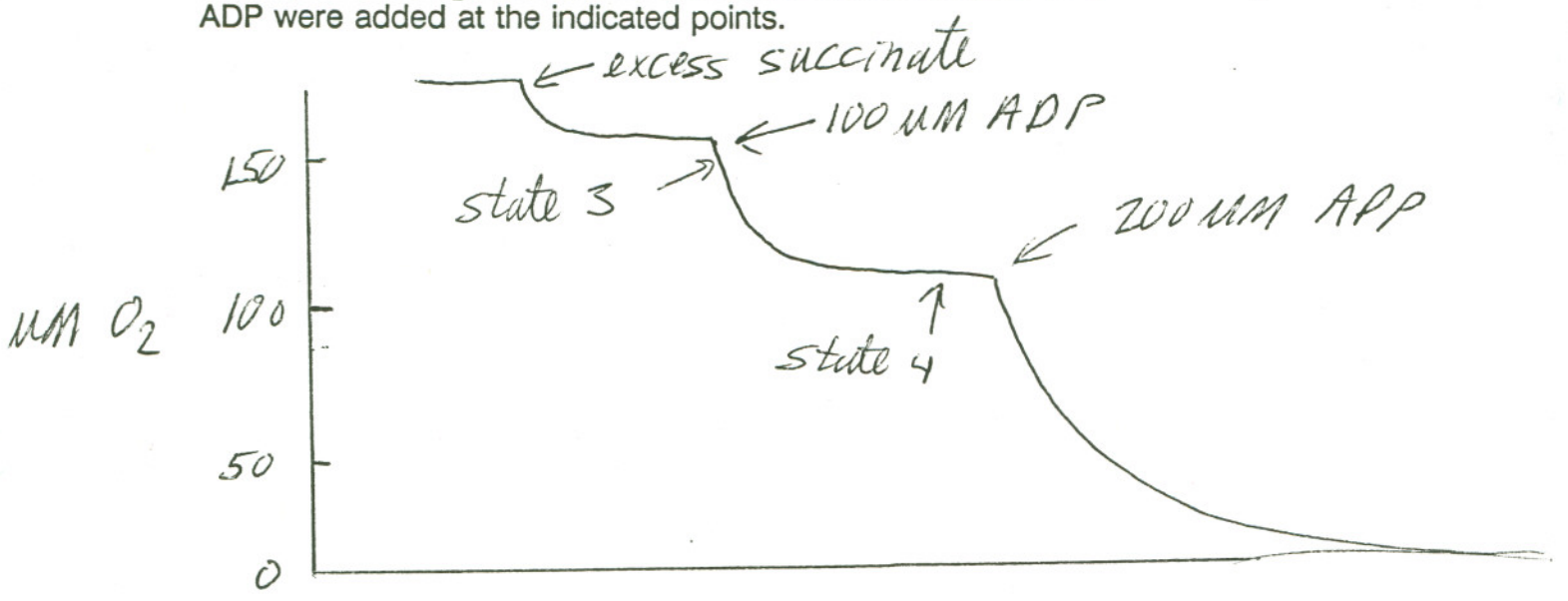
BIOCHEMISTRY CUMULATIVE EXAM

March 2, 1991

A new organism has been found that uses succinate as an energy source, and contains mitochondria that transfer electrons from succinate ($E_o' = +0.03$ V) to O_2 ($E_o' = +0.82$ V). These mitochondria contain four redox centers, A, B, C, and D, and an ATP synthase with the same properties as the one found in normal mammalian mitochondria. The following experiments were carried out on intact mitochondria isolated from this organism:

A) The mitochondria were placed in an oxygen electrode cell, excess succinate was added, and the O_2 concentration was measured as a function of time. Aliquots of ADP were added at the indicated points.

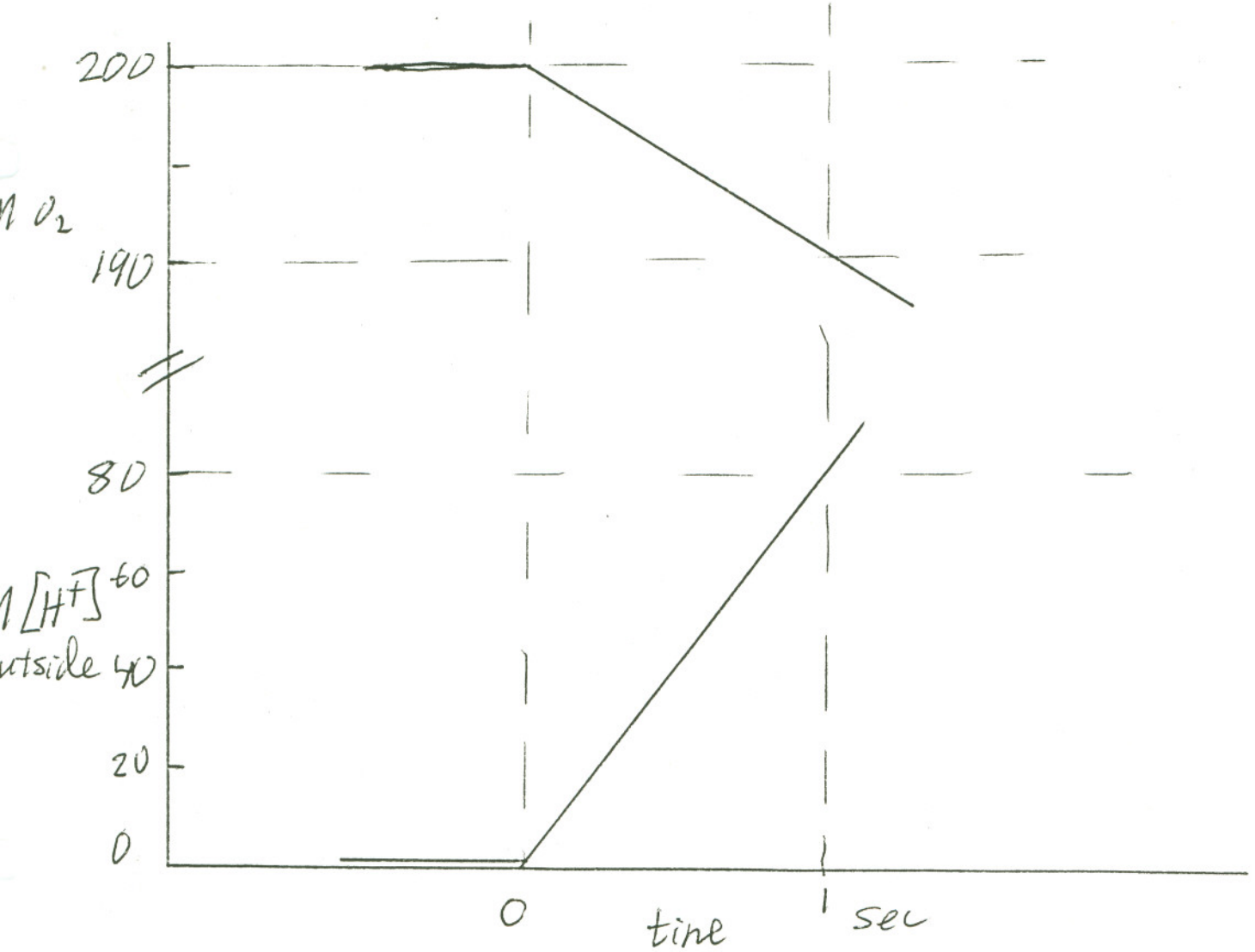


B) Spectrophotometric studies were carried out to measure the percent reduction of each redox center in states 3 and 4 identified in A) above.

Percent Reduction

	A	B	C	D
State 3:	40%	80%	20%	60%
State 4:	100%	100%	0%	100%

C) The mitochondria were added to a cell containing both an O_2 electrode and a pH electrode, and the $[O_2]$ and $[H^+]_{out}$ outside the mitochondria were measured simultaneously:



Questions

1. (20) How many ATP molecules are produced for each succinate oxidized? Show your work!

2. (20) What is the order of electron transport from succinate to O_2? Explain the reasons for your result.

3. (20) At what site(s) is(are) electron transport coupled to H^+ transport? Explain.

4. (20) How many H^+ are pumped across the membrane for each succinate oxidized? Show your work.

5. (20) Write a detailed model for the coupling between electron transport and ATP synthesis in these mitochonria.